

Statistical & Methodological Glossary

Defense Presentation Reference Sheet

Structural Limits of Training-Derived Information-Processing Systems

Jarno Matarmaa · Ural Federal University

Contents

1	Core Inferential Statistics	2
1.1	ANOVA — Analysis of Variance	2
1.2	Chi-square test (χ^2)	2
2	Effect Sizes	2
2.1	η^2 — Eta-squared (for ANOVA)	3
2.2	Cohen's d — Standardised mean difference (pairwise)	3
2.3	Cramér's V — Effect size for chi-square	4
3	ANOVA Summary Table — T6 Values Explained	4
4	Bloom's Taxonomy and the V Variable in T2	5
5	Non-Parametric Tests	5
5.1	Kruskal–Wallis Test	5
5.2	Mann–Whitney U Test	6
5.3	Bonferroni Correction — 21 pairs	6
6	Test 4 Score Distributions Figure	6
7	Category Awareness Scores — the 1–5 Scale	7
8	Mechanistic Activation-Space Comparison — the Six Points	7
8.1	Row Entropy in the Attention Heatmap	8
9	Z-Scores — Within-Test Normalisation	8
10	Novelty-Oriented Scores — How Novelty Was Measured per Test	9
11	Mechanistic Summary Values Interpreted	10
11.1	Cluster Purity = 1.0	10
11.2	Inter-Cluster Separation = 0.465	10
11.3	Entropy = 2.49	10
12	Quick Reference Table	11

Core Inferential Statistics

ANOVA — Analysis of Variance

What it is. ANOVA tests whether the *means* of three or more groups differ more than would be expected by chance. It decomposes the total variance in a dataset into *between-group* variance (driven by the factor of interest, here: AI model identity) and *within-group* variance (random noise within each model’s outputs).

The F-statistic. The test produces an F -ratio:

$$F = \frac{\text{Mean Square Between}}{\text{Mean Square Within}} = \frac{\sum_j n_j (\bar{y}_j - \bar{y})^2 / (k - 1)}{\sum_j \sum_i (y_{ij} - \bar{y}_j)^2 / (N - k)}$$

where k is the number of groups (models) and N is the total number of observations. A large F means group means are spread far apart relative to within-group scatter.

p -value. The p -value is the probability of observing an F at least as large as the obtained value *if the null hypothesis were true* (all group means are equal). Convention: $p < .05$ is called *statistically significant*; $p < .001$ is very strong evidence against the null; “n.s.” (not significant) means $p \geq .05$.

Use in the dissertation. One-way ANOVA was applied in each test family (T1–T4) with “model” (7 levels: GPT-5.2, Claude, Mistral, DeepSeek, LLaMA, Gemini, o3) as the between-subjects factor and the primary novelty-oriented metric as the dependent variable ($n = 120$ per model in T1–T3; $n \approx 17$ per model in T4).

Chi-square test (χ^2)

What it is. χ^2 tests whether the distribution of *categorical* outcomes differs across groups. In the dissertation it was used to test whether models differ in their distribution across discrete categories (e.g. Bloom taxonomy levels; transcendence vs. non-transcendence).

Formula.

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}$$

where O_i is the observed count and E_i is the expected count under the null hypothesis of independence.

Use in the dissertation. Applied in T2 (Epistemic Agency) to test whether the *proportion of paradigm-transcendent questions* varies across models.

Effect Sizes

Effect sizes quantify *how large* a difference is, independently of sample size. Statistical significance (p -value) tells you whether an effect exists; effect size tells you whether it *matters*.

η^2 — Eta-squared (for ANOVA)

Definition.

$$\eta^2 = \frac{SS_{\text{between}}}{SS_{\text{total}}}$$

η^2 is the proportion of total variance in the outcome that is *explained by the grouping variable* (here: which AI model was used).

Conventional benchmarks (Cohen 1988).

η^2 range	Label	Interpretation
≈ 0.01	Small	1% of variance explained
≈ 0.06	Medium	6% of variance explained
≥ 0.14	Large	$\geq 14\%$ of variance explained

How to read “ $\eta^2 = 0.766$ large”. $\eta^2 = 0.766$ means that **76.6% of the variance** in T4 (Category-Mistake Detection) scores is attributable to *which model produced the answer*. This is an exceptionally large effect: model identity is the dominant driver of T4 performance, far exceeding measurement noise. It justifies the conclusion that models occupy clearly distinct performance tiers on this task.

Values in this dissertation.

Test	Task	η^2	Verdict
T1	Ontological Innovation in sensory modalities	0.326	Moderate–large
T2	Epistemic Agency & question formation	0.068	Negligible (n.s.)
T3	Theory Generation (consciousness)	0.040	Negligible (n.s.)
T4	Category-Mistake Detection	0.766	Very large

T2 and T3 having negligible η^2 means all models behaved virtually identically: the training constraint is architecture-invariant. T4’s very large η^2 reflects genuine tier separation in philosophical-language fluency — but even the top tier cannot escape the training manifold.

Cohen’s d — Standardised mean difference (pairwise)

Definition.

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{\text{pooled}}}$$

d expresses the mean difference between *two* groups in units of standard deviations. Benchmarks: $|d| = 0.2$ small, 0.5 medium, 0.8 large.

Use in the dissertation. Cohen’s d was computed for every *pairwise* model comparison within each test family to accompany the Bonferroni-corrected Mann–Whitney U tests (Section 5), giving a scale-free magnitude alongside the corrected p -value.

Cramér’s V — Effect size for chi-square

What it is. Cramér’s V rescales χ^2 to lie in $[0, 1]$ regardless of table size:

$$V = \sqrt{\frac{\chi^2/N}{\min(r-1, c-1)}}$$

where r and c are the number of rows and columns in the contingency table.

Conventional benchmarks (Cramér 1946; Cohen 1988). For a 7×2 table (min = 1):

V range	Label	Meaning
≈ 0.10	Small	Weak association
≈ 0.30	Medium	Moderate association
≥ 0.50	Large	Strong association

How to read “ $V = 0.166$ small”. In T2, the χ^2 test of whether transcendence rates differ across the 7 models yielded $V = 0.166$ — a *small* effect. This means models do differ (one model, Gemini, has 0% transcendence; LLaMA drives the χ^2), but the overall association between model identity and transcendence category is weak. Combined with $\eta^2 = 0.068$ (n.s.), the conclusion is that all models share a *structurally similar low-transcendence regime*.

ANOVA Summary Table — T6 Values Explained

The slide “T6 — ANOVA Summary” reports:

Test	F -statistic	p -value	Interpretation
T1 Ontological Innovation	$F(6, 833) = 12.98$	$p < .001$	Significant model effect
T2 Epistemic Agency	$F(6, 833) = 10.09$	$p < .001$	Significant model effect
T3 Theory Generation	$F(6, 161) = 1.12$	n.s.	No model effect
T4 Category-Mistake Detection	$F(6, 112) = 88.06$	$p < .001$	Very strong model effect

Why T3 is non-significant. $F = 1.12$, n.s. means the variance *between* models’ theory-generation scores is no greater than the variance *within* models. All seven models produce theories of statistically equivalent novelty — all near zero. This is the most decisive finding: even where models differ stylistically, their semantic novelty is indistinguishable.

Why T4’s $F = 88.06$ is so large. An F of 88 means group means are nearly *88 times* more spread than within-group noise. GPT-5.2 and Mistral score dramatically higher than LLaMA and Gemini on philosophical-category recognition. However, the high η^2 here reflects fluency differences, not a constraint break.

Threshold row. “Thresholds: T1=0.40 · T2=0.62/0.67 · T3=0.30/0.70 · T4=0.40” refers to the *decision thresholds* on the primary continuous metric used to classify an output as “novelty-oriented” (above threshold) vs. “derivative” (below threshold). For T2 the two thresholds define an ambiguous middle band; for T3 a strict lower (0.30) and upper (0.70) boundary bracket the hybrid zone.

Bloom’s Taxonomy and the V Variable in T2

What is Bloom’s Taxonomy? Bloom’s Taxonomy (1956, revised 2001) is a hierarchical classification of cognitive skills, from lower-order to higher-order thinking:

Level 1. Remember — recall facts

Level 2. Understand — interpret meaning

Level 3. Apply — use knowledge in new contexts

Level 4. Analyse — break into components, identify patterns

Level 5. Evaluate — make judgements, critique

Level 6. Create — produce novel or original work

Use in T2 (Epistemic Agency). Questions generated by the models were classified along a two-level Bloom scale:

- **Exploratory** (Levels 1–4): questions that probe, analyse, or examine existing knowledge — the vast majority (91%).
- **Paradigm-challenging** (Levels 5–6): questions that contest assumptions or propose new frameworks — only $\approx 9\%$.

The near-total dominance of exploratory questions confirms that models operate as sophisticated pattern-completers, not as agents that genuinely challenge the epistemic framework they were trained on.

The $V = 0.166$ “small” label. V here is Cramér’s V (see Section 2.3), the effect size from a χ^2 test on the 7 models $\times$ 2 categories (exploratory vs. paradigm-challenging) contingency table. $V = 0.166$ is “small” by convention, meaning the cross-model variation in paradigm-challenging rates is real but modest — consistent with architecture-invariant constraint.

Non-Parametric Tests

Non-parametric tests make no assumption about the underlying distribution (they do not require normality). They are appropriate when data are ordinal or when group sizes are small.

Kruskal–Wallis Test

The non-parametric analogue of one-way ANOVA. It tests whether k groups have the same *distribution* (specifically, the same rank-sum).

$$H = \frac{12}{N(N+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} - 3(N+1)$$

Use in the dissertation. Kruskal–Wallis was applied as a robustness check alongside ANOVA (especially for T4 where score distributions are skewed), confirming that parametric findings are not artefacts of normality violations.

Mann–Whitney U Test

Pairwise non-parametric comparison of two groups based on rank sums — the non-parametric analogue of the independent-samples t -test.

Use in the dissertation. Applied to all $\binom{7}{2} = 21$ model pairs within each test to identify which specific pairs of models differ significantly.

Bonferroni Correction — 21 pairs

The multiple-comparisons problem. If you run 21 independent tests at $\alpha = .05$, the probability of getting *at least one* false positive by chance is $1 - (0.95)^{21} \approx 0.66$ — far above 5%.

Bonferroni correction. Divide the desired α by the number of tests:

$$\alpha_{\text{corrected}} = \frac{\alpha}{m} = \frac{.05}{21} \approx .0024$$

Only pairs with $p < .0024$ are declared significant. This is conservative but controls the *family-wise error rate* — the probability of any false positive across the entire 21-pair family.

“21 pairs” comes from choosing 2 models from 7: $\binom{7}{2} = 21$.

Test 4 Score Distributions Figure

What Test 4 measures. T4 (Category-Mistake Detection) assessed models on three philosophical sub-tasks drawn from Gilbert Ryle’s category-mistake framework:

1. **Panel (a) — Distinction Recognition** ($\bar{x} = 2.58$): Can the model identify that two concepts belong to different ontological categories (e.g. “mind” vs. “brain”)?
2. **Panel (b) — Mistake Identification** ($\bar{x} = 2.79$): Given a sentence containing a category error (assigning a predicate to the wrong ontological type), can the model flag it and explain why it is an error?
3. **Panel (c) — Framework Proposal** ($\bar{x} = 3.31$): Can the model propose a corrected ontological framework that avoids the error? This is the most generative sub-task and yields the highest mean score.

Scores are on a **1–5 ordinal scale** (see Section 7). The distributions show that models cluster at middle values for (a) and (b) but shift upward for (c), suggesting stronger pattern-matching to repair-style text than to detect or explain errors.

“Ceiling” in the dissertation. “Ceiling” refers to the theoretical maximum score (5 on the rubric, or 100% on a binary metric). *Ceiling effects* occur when many responses cluster near the maximum, compressing variance and making it hard to distinguish the best performers. In the presentation, “mathematically grounded creativity ceiling” is used metaphorically: training-derived manifolds impose a hard upper bound on ontological novelty, just as a score ceiling caps measurement.

Category Awareness Scores — the 1–5 Scale

What the scale measures. T4 responses were evaluated by a structured rubric on a **5-point ordinal scale** (1–5, higher = better awareness of philosophical categories):

Score	Meaning
1	No category awareness; the response treats the ontological distinction as irrelevant or conflates categories.
2	Partial awareness; the model hints at a distinction but does not articulate it.
3	Adequate awareness; correct identification with limited explanation.
4	Good awareness; clear explanation with appropriate philosophical vocabulary.
5	Full awareness; precise identification, detailed explanation, and correct corrective framework proposed.

So **5 = fully category-aware** (philosophically competent), and **1 = totally unaware** (treats the error as unproblematic). Zero is not used — the scale begins at 1.

Model-level means. GPT-5.2 $\bar{x} = 0.825$ (normalised to $[0, 1]$) and Mistral $\bar{x} = 0.758$ represent the top tier; LLaMA $\bar{x} = 0.381$ the bottom. When reported as a proportion rather than a raw mean, 0.825 corresponds to a raw mean of approximately 4.1 on the 1–5 scale.

Importantly, even the top-tier models achieve these high T4 scores by exploiting *philosophical-language patterns seen in training data* — the high traceability (95.2% across all models) confirms that the responses reproduce known frameworks rather than constructing new ones.

Mechanistic Activation-Space Comparison — the Six Points

What the figure shows. The “Mechanistic activation-space comparison” figure (T5) plots the *internal activation vectors* of DistilBERT (used as a mechanistic proxy) in a 2-D space produced by PCA followed by t-SNE dimensionality reduction. Each point in the figure represents the *mean activation centroid* for one experimental condition.

Why six points? The six points correspond to the **four test categories** (T1–T4) plus **two baseline conditions** (structurally novel input; training-matched control), giving $4 + 2 = 6$ distinct condition centroids. The rationale: if internal representations are undifferentiated, all six points would cluster together; structured separation indicates the model encodes different task types differently in activation space.

What the separation tells us. The partial-but-structured separation observed (inter-cluster distance = 0.465, purity = 1.0) means:

- Each condition lands in a distinct region of activation space (purity = 1.0: no overlap between clusters).
- The distance between clusters (0.465 on the cosine similarity scale) is moderate — they are separated but not maximally distant.

- Crucially, all six centroids remain *within* the convex hull of training-modality representations — confirming that even internal processing geometry is bounded by the training manifold.

Row Entropy in the Attention Heatmap

What it is. In a transformer’s self-attention mechanism, each token attends to every other token with a weight $a_{ij} \geq 0$, $\sum_j a_{ij} = 1$. The *row entropy* for query position i is the Shannon entropy of that attention distribution:

$$H_i = - \sum_j a_{ij} \log_2 a_{ij}$$

Interpretation.

- **High row entropy** (H_i close to $\log_2 n$, where n is sequence length): the attention is *diffuse* — the token attends roughly equally to all positions. The head is not performing sharp, task-specific lookup.
- **Low row entropy** (H_i close to 0): the attention is *focused* — the token strongly attends to one or a few specific positions. This indicates structured pattern retrieval.

In the T5 heatmap (panels a–b: token-level attention; panel c: row entropy and max-attention trends), the entropy profile reveals how “deliberate” each attention head’s behaviour is across query positions. High entropy across all heads would indicate the model is not exploiting internal structure; the observed mixed profile (some heads focused, some diffuse) is consistent with *training-derived feature extraction* rather than open-ended exploration.

Z-Scores — Within-Test Normalisation

What a z-score is. A z-score transforms a raw score x into units of standard deviation relative to a reference distribution:

$$z = \frac{x - \bar{x}}{s}$$

where $\bar{x}$ is the mean and s is the standard deviation of the group.

Interpretation.

- $z = 0$: exactly average performance within the test.
- $z > 0$ (e.g. $z = +1.5$): the model performed **1.5 standard deviations above** the cross-model mean on this test — substantially better than typical.
- $z < 0$ (e.g. $z = -1.2$): **1.2 standard deviations below** the mean — notably worse than typical.

Use in the dissertation. Z-scores were computed *within each test* to allow *cross-test comparison* of model profiles (“Cross-test model comparison heatmaps” on Slide 14). Because T1–T4 use different metrics and scales, raw scores cannot be directly compared; z-scores place each model’s performance in a common framework. A model with $z = +2$ on T4 but $z = -0.5$ on T1 is strong at philosophical-language fluency but weaker

at generating ontologically novel proposals — a profile that reveals what each model’s training prioritised.

The model performance heatmap (“Within-test z-scores cross-model”) visualises these z-scores as a colour gradient; cells far from zero (dark blue/red) indicate outlier performance relative to the 7-model cohort.

Novelty-Oriented Scores — How Novelty Was Measured per Test

Common idea. Across all tests, a “novelty-oriented” score estimates *how far an output diverges from known training patterns*. The exact operationalisation differs by test because each task probes novelty at a different level of representation.

Test	How novelty was measured
T1 Ontological Innovation	Cosine similarity between the embedding of a proposed sensory modality and the nearest training-modality centroid (8 modalities: vision, audition, ...). Low similarity = high novelty. Threshold: $\text{sim} < 0.40$ to enter the “novelty-positive” zone. Novelty band assignment (near/moderate/far/missing) was also used.
T2 Epistemic Agency	Binary classification: does the generated question <i>transcend</i> the training-derived conceptual framework (paradigm-challenging) or remain within it (exploratory)? Transcendence rate ($\approx 5\%$ across models) is the primary novelty metric. Traceability to the literature was also scored.
T3 Theory Generation	Semantic similarity to a corpus of 46 known consciousness theory families (sentence-transformer embeddings). A three-stage CF-detection algorithm flagged purely computational-functionalist outputs. A “gated novel” count required similarity below thresholds (0.30 lower / 0.70 upper) to pass all three gates simultaneously. Result: 0 theories achieved framework-external novelty.
T4 Category Recognition	A 1–5 rubric score (see Section 7). Novelty-orientation here means the ability to detect category violations and propose ontologically distinct corrective frameworks — a more local, task-specific form of novelty. Scores ≥ 4 were operationalised as “novelty-oriented” for the purpose of cross-test comparison.

Key difference between tests. T1–T3 measure *output-level* semantic novelty (does the text escape known conceptual territory?); T4 measures *meta-level* novelty awareness (does the model recognise and repair violations of ontological categories?). The contrast is deliberate: even high T4 scores do not imply the model can *produce* genuinely novel frameworks — only that it can *recognise* philosophical-language patterns well enough to discuss category errors.

Mechanistic Summary Values Interpreted

The final summary bullet on Slide 15 reads:

“Mechanistic: cluster purity 1.0, inter-cluster separation 0.465, entropy 2.49 — internal representations fully stratified by training-derived structure.”

Cluster Purity = 1.0

k -means clustering ($k = 4$) was applied to the activation vectors of DistilBERT across all 6 experimental conditions. *Cluster purity* is defined as:

$$\text{Purity} = \frac{1}{N} \sum_j \max_c |C_j \cap L_c|$$

where C_j is the set of points in cluster j and L_c is the set of points with true label c . **Purity = 1.0** means every cluster contains only points from a single condition — perfect separation. The model internally distinguishes all experimental conditions with zero confusion.

Inter-Cluster Separation = 0.465

This is the *mean pairwise cosine distance* between cluster centroids. Cosine distance ranges from 0 (identical direction) to 2 (opposite directions); 0.465 indicates *moderate separation* — clusters are clearly distinct but not maximally different.

Entropy = 2.49

This is the *Shannon entropy* of the overall attention weight distribution averaged across all heads and positions:

$$H = - \sum_i p_i \log_2 p_i$$

How to interpret the value 2.49 bits. The theoretical maximum entropy for a uniform distribution over n positions is $\log_2 n$. For a typical sequence of length ≈ 50 tokens, $H_{\max} = \log_2 50 \approx 5.64$ bits. An entropy of 2.49 is roughly 44% of the maximum, indicating that attention is *neither maximally diffuse nor maximally focused* — a middle ground consistent with structured feature extraction from training-derived representations.

In context: a random, untrained model would show entropy close to $H_{\max}$ (no focus); a model that only looks at one token would show entropy near 0. Entropy = 2.49 confirms that the internal processing is **organized and structured** — but that structure mirrors training-data patterns, not emergent open-ended computation.

The phrase *“internal representations fully stratified by training-derived structure”* summarises all three mechanistic metrics together: the activation space is partitioned cleanly (purity 1.0), the partitions are meaningfully separated (0.465), and the attention mechanism operates with intermediate, structured focus (2.49 bits) — all consistent with a model that has internalized training-derived categories deeply but has not transcended them.

Quick Reference Table

Symbol / Term	One-line meaning
F	Ratio of between-group to within-group variance (ANOVA)
p	Probability of observed result under null hypothesis
n.s.	Not significant ($p \geq .05$)
η^2	Proportion of variance explained by the group factor
Cohen’s d	Mean difference in standard deviation units (2 groups)
Cramér’s V	Effect size for χ^2 ; association strength in $[0, 1]$
χ^2	Test of independence between categorical variables
Kruskal–Wallis	Non-parametric ANOVA on ranks
Mann–Whitney U	Non-parametric pairwise comparison on ranks
Bonferroni	Divide α by number of tests to control false positives
z-score	Standard deviation units from the within-test mean
Row entropy	Shannon entropy of one attention-head row ($-\sum a_{ij} \log a_{ij}$)
Cluster purity	Fraction of points correctly assigned by k -means
$\eta^2 = 0.766$	76.6% of T4 variance explained by model — very large effect
$V = 0.166$	Small cross-model effect on transcendence rate (T2)
$H = 2.49$	Moderate, structured attention focus ($\approx 44\%$ of max)
Ceiling (score)	Maximum achievable on a rubric; ceiling effects compress high end
Creativity ceiling	Theoretical bound on ontological novelty imposed by training
Bloom Taxonomy	6-level hierarchy of cognitive skills (remember $\rightarrow$ create)

Prepared as a companion reference for the defense presentation of
Structural Limits of Training-Derived Information-Processing Systems.
 Jarno Matarmaa, Ural Federal University, 2026.